WORKED SOLUTION

STEP III 2007, Question 1

Pure • Trigonometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q1

The key idea

The tangent of a sum of angles is a ratio of *symmetric* functions of the individual tangents, and symmetric functions of the roots of a polynomial are exactly what Vieta's formulae hand us in terms of the coefficients. For the final part, the trick is to turn the trigonometric equation into a quartic in $t = \tan\theta$; the symmetry of its coefficients then forces the tangent of the sum to vanish.

Solution

The tangent of a sum of four angles

Split the sum as $A + B$ with $A = \theta_1 + \theta_2$ and $B = \theta_3 + \theta_4$, so that

$$\tan A = \frac{t_1 + t_2}{1 - t_1 t_2}, \qquad \tan B = \frac{t_3 + t_4}{1 - t_3 t_4}.$$

Then, multiplying the top and bottom of $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$ by $(1 - t_1 t_2)(1 - t_3 t_4)$,

$$\tan(A+B) = \frac{(t_1 + t_2)(1 - t_3 t_4) + (t_3 + t_4)(1 - t_1 t_2)}{(1 - t_1 t_2)(1 - t_3 t_4) - (t_1 + t_2)(t_3 + t_4)}.$$

The numerator is $t_1 + t_2 + t_3 + t_4 - (t_1 t_3 t_4 + t_2 t_3 t_4 + t_1 t_2 t_3 + t_1 t_2 t_4)$, and the denominator is $1 - t_1 t_2 - t_3 t_4 + t_1 t_2 t_3 t_4 - (t_1 t_3 + t_1 t_4 + t_2 t_3 + t_2 t_4)$. Writing

$$\sigma_1 = \sum t_i, \quad \sigma_2 = \sum_{i<j} t_i t_j, \quad \sigma_3 = \sum_{i<j<k} t_i t_j t_k, \quad \sigma_4 = t_1 t_2 t_3 t_4$$

for the elementary symmetric functions, this is

$$\boxed{\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{\sigma_1 - \sigma_3}{1 - \sigma_2 + \sigma_4}.}$$

(The pattern is the same as for two or three angles: odd symmetric functions with alternating signs on top, even ones on the bottom.)

In terms of the coefficients

If $t_1, \ldots, t_4$ are the roots of $at^4 + bt^3 + ct^2 + dt + e = 0$, then $at^4 + bt^3 + ct^2 + dt + e = a(t - t_1)(t - t_2)(t - t_3)(t - t_4)$, and comparing coefficients (Vieta) gives

$$\sigma_1 = -\frac{b}{a}, \qquad \sigma_2 = \frac{c}{a}, \qquad \sigma_3 = -\frac{d}{a}, \qquad \sigma_4 = \frac{e}{a}.$$

Substituting and multiplying top and bottom by $a \neq 0$,

$$\boxed{\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{d - b}{a - c + e}.}$$

The final part

Since $1 + \cos 2\theta = 2\cos^2\theta$, the equation is

$$2p\cos^2\theta + \cos\theta\cos\alpha + \sin\theta\sin\alpha = 0.$$

Dividing by $\cos^2\theta$ (we ignore the special case $\cos\theta = 0$) gives

$$2p = -(\cos\alpha + t\sin\alpha)\sec\theta, \qquad t = \tan\theta.$$

The $\sec\theta$ is not a rational function of t, so we square, using $\sec^2\theta = 1 + t^2$:

$$4p^2 = (\cos\alpha + t\sin\alpha)^2(1 + t^2).$$

Expanding,

$$\sin^2\alpha\, t^4 + 2\sin\alpha\cos\alpha\, t^3 + t^2 + 2\sin\alpha\cos\alpha\, t + (\cos^2\alpha - 4p^2) = 0, \qquad (*)$$

using $\sin^2\alpha + \cos^2\alpha = 1$ for the t^2 coefficient.

Ask yourself Squaring can introduce extra solutions. Are the $\tan\theta_i$ really the four roots of $(*)$?

Squaring means $(*)$ is also satisfied by the solutions of $p\cos 2\theta - \cos(\theta - \alpha) + p = 0$. But replacing θ by $\theta + \pi$ leaves $t = \tan\theta$ and $\cos^2\theta$ unchanged and changes the sign of $\cos(\theta - \alpha)$. So for each real root t of $(*)$, of the two angles in $[0, 2\pi)$ with tangent t, exactly one satisfies the original equation (not both, since that would need $\cos(\theta - \alpha) = 0$ and hence $p\cos^2\theta = 0$, a special case). Thus the solutions in $[0, 2\pi)$ correspond one-to-one with the real roots of $(*)$, and the four given angles have as their tangents the four roots of $(*)$.

Now apply the previous result with $a = \sin^2\alpha$, $b = d = 2\sin\alpha\cos\alpha$, $c = 1$, $e = \cos^2\alpha - 4p^2$ (we need $a \neq 0$, i.e. $\sin\alpha \neq 0$; otherwise $(*)$ is not a quartic, another special case). The numerator is $d - b = 0$, and the denominator is

$$a - c + e = \sin^2\alpha - 1 + \cos^2\alpha - 4p^2 = -4p^2,$$

which is non-zero (if $p = 0$ the equation is just $\cos(\theta - \alpha) = 0$, with only two solutions). Hence

$$\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0,$$

so $\theta_1 + \theta_2 + \theta_3 + \theta_4 = n\pi$ for some integer n, as required.

Checks

- Put $t_3 = t_4 = 0$ in the four-angle formula: it collapses to $(t_1 + t_2)/(1 - t_1 t_2)$, the familiar two-angle formula.
- With $p = \frac{1}{2}$ and $\alpha = 3$, solving the equation numerically on $[0, 2\pi)$ gives four roots whose sum, divided by π, is an integer to within 10^{-9}. (Not every p and α give four solutions: $p = \frac{1}{2}$, $\alpha = 1$ gives only two.) The same held for every random (p, α) we tried that gave four solutions, and we also checked both formulae on thousands of random angles and quartics.

Common slips

- Sign errors in Vieta: the alternate coefficients b and d come with minus signs.
- Trying to write $\cos(\theta - \alpha)$ directly in terms of $\tan\theta$ without squaring, or squaring without checking that no unwanted angles slip in.

Beyond STEP

The expression of tan of a sum through elementary symmetric functions is one instance of a big theorem: every symmetric polynomial in the roots is a polynomial in the coefficients. It is also neatly explained by complex numbers, since $\prod(1 + \mathrm{i}t_k) = \prod \sec\theta_k \, \mathrm{e}^{\mathrm{i}\sum\theta_k}$, and its real and imaginary parts are $1 - \sigma_2 + \sigma_4$ and $\sigma_1 - \sigma_3$. Needham's *Visual Complex Analysis* develops this geometric way of thinking, and Siklos's *Advanced Problems in Mathematics* has more STEP problems in the same spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.